QUANTUM INVARIANTS OF MANIFOLDS

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1. Equivariant index formula for circle action

1.1. **Atiyah-Singer index theorem.** Arguably this is the best theorem in the past century.

Theorem 1.1 (Atiyah-Singer)**.** *Let X be a closed oriented smooth manifold of dimension $2n$ and $D = (D_i : \Gamma E_i \to \Gamma E_{i+1})$ an elliptic complex $(i = 0, \dots, m-1)$, associated to the tangent bundle. Then the index of this complex is determined by the following formula*

$$\mathrm{ind}(D) = (-1)^n \left(\Big(\frac{1}{e(T^*X)} \sum_{i=0}^{m} (-1)^i ch(E_i) \Big) \cdot td(TX \otimes \mathbb{C}) \right) [X].$$

Recall that $\mathrm{ind}(D)$ is defined to be the Euler characteristic of the elliptic complex D. Since each D_i is elliptic, $\mathrm{coker}(D_i)$ is finite dimensional, so Euler characteristic is well-defined. The proof of Atiyah-Singer index theorem is trivial analysis plus trivial topology. The topological essence of index theorem is the formula

$$ch(U_K) = \pm U_H / Td$$

which we established last time.

For convenience, if we write $x_1, \dots x_n$ to be the formal Chern roots of TX, then

$$\mathrm{ind}(D) = \left(\Big(\sum_{i=0}^{m} (-1)^i ch(E_i) \Big) \Big(\prod_{j=1}^{n} \frac{x_j}{1 - e^{-x_j}} \cdot \frac{1}{1 - e^{x_j}} \Big) \right) [X].$$

One has to use this formula with caution, since the formula is neither symmetric in x_j^2 nor an invertible power series.

Now let's discuss several important examples.

1.2. **The de Rham complex.** Let X be compact smooth of dimension k and T its tangent bundle. Let $E_i = \Lambda^i(T^* \otimes \mathbb{C})$. The exterior derivative d yields an elliptic complex. From definition, $\mathrm{ind}(d) = \sum_i \dim_{\mathbb{C}} H^i_{dR} = e(X)$. On the other hand, we have

$$ch\Big(\sum_{i=0}^{n} (\Lambda^i(T^* \otimes \mathbb{C})) \cdot y^i \Big) = \prod_{j=1}^{n} \Big((1 + ye^{x_j})(1 + ye^{-x_j}) \Big).$$

1.3. **The Dolbeault complex.** Let X be a complex n-dimensional manifold, T the holomorphic tangent bundle, *i.e.* $T = T^{0,1}$. With notation $A^{p,q} = \Gamma(\Lambda^p T^* \otimes \Lambda^q \overline{T}^*)$. The index of $\bar{\partial}$ for fixed p is denoted by χ^p, namely

$$\chi^p = \sum_{q=0}^{n} (-1)^q \cdot h^{p,q}.$$

It then follows from index formula that

$$\chi_y := \sum_{p=0}^{n} \chi^p \cdot y^p = \prod_{j=1}^{n} \Big((1 + ye^{-x_j}) \frac{x_j}{1 - e^{-x_j}} \Big)[X].$$

If we replace x by $x(1+y)$ then an easy exercise shows χ_y is the genus belonging to the power series

$$Q(x) = \frac{x(1 + ye^{-x(1+y)})}{1 - e^{-x(1+y)}}.$$

In particular, χ_0 is Todd genus or sometimes called arithmetic genus, χ_{-1} is the Euler characteristic, and χ_1 is the signature.

1.4. **The signature as an index.** Let X be compact oriented $4k$-dimensional smooth manifold, let T be its tangent bundle provided with a Riemannian metric. Recall that Hodge star operator has the property that $\star^2 = (-1)^{i(n-i)} \cdot \mathrm{id}$ on $\Gamma^i(T^* \otimes \mathbb{C})$. For this, we define $\tau = (-1)^{i(i-1)/2+k}\star$, then $\tau^2 = \mathrm{id}$ holds. Hence τ decompose $\Lambda(T^* \otimes \mathbb{C})$ into direct sum of sub-bundles E_+ with eigenvalue $+1$ and E_- with eigenvalue -1.

Note that $\tau(d + d^*) = -(d + d^*)\tau$, one can consider the elliptic operator $d + d^* : \Gamma E_+ \to \Gamma E_-$. One can show

$$ch(E_+) - ch(E_-) = \prod_{j=1}^{2k} (e^{x_j} - e^{-x_j}).$$

It then follows that $\mathrm{ind}(d + d^*) = \mathrm{sign}(X)$.

1.5. **The equivariant index.** Let X be a compact complex manifold of complex dimension n, and as before $D = (D_i : \Gamma E_i \to \Gamma E_{i+1})$ be an elliptic complex. Let G be a compact topological group acting on X by holomorphic maps. In addition, assume G acts on the elliptic complex, *i.e.* G also acts on the bundle E_i (e.g. if all bundles E_i are associated to tangent bundle of X) and this action commutes with the differential operators D_i. Then G also acts on the cohomology groups H^i of the complex.

Definition 1.2. *The equivariant index* $\mathrm{ind}(g, D)$ *of* D *is defined to be*

$$\mathrm{ind}(g, D) := \sum_{i=0}^{m} (-1)^i \cdot \mathrm{tr}(g, H^i).$$

Example 1.3. $\mathrm{ind}(D) = \mathrm{ind}(id, D)$.

Now we want to compute equivariant index using topological information of X and the group action. Let $X^g = x \in X : g \cdot x = x$ denote the fixed point set of g. This is a complex submanifold of X which is not necessarily connected. We furthermore decompose $X = \cup X^g_\nu$ into connected components. The X^g_ν are connected submanifolds of X of possibly different dimensions.

Let's fix such a component $Y = X^g_\nu$. For a point $p \in Y$, g acts linearly on the tangent space T_pX. There exists a Hermitian metric on $TX|_Y$ so that g acts unitarily on $TX|_Y$. Therefore T_pX decomposes into the direct sum of eigenspaces $N_{p,\lambda}$ for eigenvalues λ of modulus one. Since G acts continuously on X, one further obtains an eigen bundle N_λ over Y. Indeed, N_1 is precisely the tangent bundle TY of Y. Under the variation of points p in Y, the eigenvalues cannot change since depend only on the isomorphism type of representation on T_pX of the subgroup generated

by g, and representation ring of this subgroup is discrete. With $d_\lambda = \mathrm{rk} N_\lambda$ we therefore have:

$$TX|_Y = \oplus_\lambda N_\lambda, \quad c(N_\lambda) = \prod_{i=1}^{d_\lambda}(1 + x_i^{(\lambda)}).$$

Now we are in position to write down an index formula for $\mathrm{ind}(g, D)$, as a sum of contributions $a(X_\nu^g)$ corresponding to each components of X_ν^g. We recall the original index formula

$$\mathrm{ind}(D) = \left(\Big(\sum_{i=0}^{m} (-1)^i ch(E_i) \Big) e(X) \Big(\prod_{j=1}^{n} \frac{1}{1 - e^{-x_j}} \cdot \frac{1}{1 - e^{x_j}} \Big) \right) [X].$$

To obtain the contribution from $Y = X_\nu^g$, we replace, in the above formula, X by Y, e^{x_i} by $\lambda^{-1} \cdot e^{x_i}$ and e^{-x_i} by $\lambda \cdot e^{-x_i}$ (where x_i) belongs to the eigenvalue λ. We apply the same process to the terms $ch(E_i)$. This can obviously be done if the E_i are associated to the tangent bundle of X. For $\lambda = 1$, the term $1 - \lambda^{-1} \cdot e^{x_i}$ is not invertibe; but those x_i, which belong to the eigenvalue 1, originate from the tangent bundle of Y and so cancel with the factor $e(Y)$.

1.6. **The equivariant χ_y-genus for S^1-actions.** Applying the recipe to the χ_y-genus, we obtain from

$$\chi_y(X) = \sum_{p=0}^{n} \chi^p(X) \cdot y^p = \prod_{i=1}^{n} \Big((1 + ye^{-x_i}) \frac{x_i}{1 - e^{-x_i}} \Big) [X]$$

the equivariant formula

$$\chi_y(g, X) = \sum_{p=0}^{n} \chi^p(g, X) \cdot y^p = \sum_\nu a(X_\nu^g),$$

where we have for each fixed point component $Y = X_\nu^g$:

$$a(Y) = \Big(\prod_{\lambda \neq 1} \prod_{i=1}^{d_\lambda} \big(\frac{1 + y \cdot \lambda e^{-x_i^{(\lambda)}}}{1 - \lambda e^{-x_i^{(\lambda)}}} \big) \cdot \prod_{i=1}^{d_1} \big(1 + ye^{-x_i^{(1)}} \big) \frac{x_i^{(1)}}{1 - e^{-x_i^{(1)}}} \Big) [Y].$$

Now let $G = S^1$ and let $q \in S^1$ be a topological generator. Then we have $X^q = X^{S^1}$. In this situation, the whole group S^1 acts on the restriction of the tangent bundle of X to X^q. Recall if S^1 acts on a vector space, then one writes

$$V = \sum_{k \in \mathbb{Z}} q^k V_k$$

which means that $V = \oplus_k V_k$ and q acts on V_k as multiplication by q^k. This way, we have a splitting

$$TX|_Y = \sum_{k=-\infty}^{\infty} q^k N_k.$$

The eigenvalues λ of q are now all integral powers q^k of q, the equivariant χ_y-genus is then ($d_k = \mathrm{rk} N_k$)

$$\chi_y(q, X) = \sum_\nu a(X_\nu^{S^1})$$

with

$$a(X_\nu^{S^1}) = \Big(\prod_{i=1}^{d_0} x_i^{(1)} \frac{1+ye^{-x_i^{(1)}}}{1-e^{-x_i^{(1)}}} \prod_{k\neq 0}\prod_{j=1}^{d_k} \big(\frac{1+y\cdot q^k e^{-x_j^{(q^k)}}}{1-q^k e^{-x_j^{(q^k)}}}\big)\Big)[X_\nu^{S^1}].$$

Note that $\chi_y(q,X)$ is a finite Laurent series in q, and it is easy to see

$$a(X_\nu^{S^1})_{q=0} = \chi_y(X^{S^1_\nu})\cdot(-y)^{\sum_{k<0} d_k}, \quad a(X_\nu^{S^1})_{q=\infty} = \chi_y(X_\nu^{S^1})\cdot(-y)^{\sum_{k>0} d_k}.$$

Hence $\chi_y(q,X)$ is constant in q, and $\chi_y(q,X)\equiv\chi_y(id,X)=\chi_y(X)$.

1.7. **The equivariant signature for S^1-actions.** Now let X be a $2n$-dimensional compact, oriented, differentiable manifold with S^1-action. Note that TX is a real vector bundle, we have splitting

$$TX|_{X_\nu^{S^1}} = \sum_{k\geq 0} q^k N_k = TX_\nu^{S^1} \oplus \sum_{k>0} q^k N_k.$$

The bundle N_k are real bundles, which for $k>0$ obtain a complex structure through the identification with a complex vector bundle. This identification yields a unique orientation on N_k, in view of the condition $k>0$. We orient $N_0 = TX_\nu^{S^1}$ so that all orientations taken together yield the orientation of X.

Having made this splitting, the equivariant index theorem yields $\mathrm{sign}(q,X) = \sum_\nu a(X_\nu^{S^1})$ with

$$a(X_\nu^{S^1}) = \Big(\prod_{i=1}^{d_0} x_i^{(1)} \frac{1+e^{-x_i^{(1)}}}{1-e^{-x_i^{(1)}}} \prod_{k>0}\prod_{j=1}^{d_k} \big(\frac{1+q^k e^{-x_j^{(q^k)}}}{1-q^k e^{-x_j^{(q^k)}}}\big)\Big)[X_\nu^{S^1}].$$

Since S^1 is connected, it acts trivially on cohomology, therefore the equivariant signature does not depend on q. We can put $q=0$ and have

$$\mathrm{sign}(X) = \sum_\nu \mathrm{sign}(X_\nu^{S^1}).$$

2. Quantization of classical genera

Let X be a compact, oriented manifold of dimension $4k$. The the free loop space of X is the infinite dimensional manifold

$$\mathcal{L}X = \{g : S^1 \to X : g \text{ differentiable}\}.$$

There is a canonical action of S^1 on $\mathcal{L}X$, with $(\mathcal{L}X)^{S^1} = X$ as constant loops. It is not hard to show that at a constant loop $g\equiv p\in X$, $T_p(\mathcal{L}X)\simeq\mathcal{L}(T_pX)$ where the S^1 acts canonically on $\mathcal{L}(T_pX)$. We shall denote $TX\otimes\mathbb{C}$ by $T_\mathbb{C}$, we have a splitting

$$T(\mathcal{L}X)|_X = TX \oplus \sum_{n>0} q^n T_\mathbb{C}.$$

2.1. **Quantization of signature-the construction.** With the standard notation

$$p(X) = 1+p_1+\cdots+p_k = (1+x_1^2)\cdots(1+x_{2k}^2)$$
$$c(T_\mathbb{C}) = (1+x_1)\cdots(1+x_{2k})(1-x_1)\cdots(1-x_{2k})$$

The recipe of the S^1-equivariant signature formula on free loop space yields

Definition 2.1. *The quantum signature of X is*

$$\operatorname{sign}(q, \mathcal{L}X) := \prod_{i=1}^{2k} \Big(x_i \frac{1+e^{-x_i}}{1-e^{-x_i}} \cdot \prod_{n=1}^{\infty} \Big(\frac{(1+q^n e^{-x_i})(1+q^n e^{x_i})}{(1-q^n e^{-x_i})(1-q^n e^{x_i})} \Big) \Big)[X].$$

This power series is symmetric in x_i^2, and after evaluation on X is a power series in q with rational coefficients. Recall that

Definition 2.2 (Twisted signature)**.** *Let X be an oriented manifold of dimension $2k$ and W a complex vector bundle over X. Then the signature of X with values in the vector bundle W is defined as*

$$\operatorname{sign}(X, W) := \Big(\prod_{i=1}^{k} x_i \frac{1+e^{-x_i}}{1-e^{-x_i}} \cdot ch(W) \Big)[X].$$

Definition 2.3 (Twisted $\hat{A}$-genus)**.** *Let X be an oriented manifold of dimension $2k$ and W a complex vector bundle over X. Then the $\hat{A}$-genus of X with values in the vector bundle W is defined as*

$$\hat{A}(X, W) := \Big(\prod_{i=1}^{k} \frac{x_i/2}{\sinh(x_i/2)} \cdot ch(W) \Big)[X].$$

Then with an practiced eye, one observes

Theorem 2.4. *We have*

$$\operatorname{sign}(q, \mathcal{L}X) = \operatorname{sign}(X, \bigotimes_{n=1}^{\infty} S_{q^n} T_{\mathbb{C}} \otimes \bigotimes_{n=1}^{\infty} \Lambda_{q^n} T_{\mathbb{C}}).$$

Therefore $\operatorname{sign}(q, \mathcal{L}X)$ *is a power series in q with integral coefficients and constant term (the coefficient of q^0)* $\operatorname{sign}(X)$.

It is well-known that a product $\prod(1+u_i)$ converges absolutely provided the series $\sum |u_i|$ converges, hence in our case only for $|q| < 1$. We have therefore defined quantum signature formally, but it is meaningful as a power series.

2.2. **Quantization of signature-modularity.** Notice that quantum signature is not normalized, in the sense that it has constant term

$$\Big(2 \cdot \prod_{n=1}^{\infty} \frac{(1+q^n)^2}{(1-q^n)^2} \Big)^{2k}.$$

Lemma 2.5. *The infinite product*

$$\Big(2 \cdot \prod_{n=1}^{\infty} \frac{(1+q^n)^2}{(1-q^n)^2} \Big)^{-4}$$

is a modular form of weight 4 on $\Gamma_0(2)$, and is indeed precisely our modular form ε.

Thus we can rewrite the quantum signature as

$$\operatorname{sign}(q, \mathcal{L}X) = \varphi(X) \cdot \varepsilon^{-k/2}$$

where $\varphi(X)$ is the genus associated to

$$\frac{x}{f(x)} = \frac{1}{2} \frac{x}{\tanh(x/2)} \cdot \prod_{n=1}^{\infty} \Big(\frac{(1+q^n e^{-x_i})(1+q^n e^{x_i})}{(1-q^n e^{-x_i})(1-q^n e^{x_i})} : \frac{(1+q^n)^2}{(1-q^n)^2} \Big).$$

It turns out this is precisely the expansion of $x\sqrt{\wp(x)-e_1}$. Recall that a genus is elliptic if and only if $f(x)=x/Q(x)$ satisfies $(f')^2=1-2\delta f^2+\varepsilon f^4$. One can then check

Corollary 2.6. *$\varphi(X)$ is an elliptic genus of X, hence a modular form of weight $2k$ on $\Gamma_0(2)$. The quantum signature $\operatorname{sign}(q,\mathcal{L}X)=\varphi(X)\cdot\varepsilon^{-k/2}$ is a modular function on $\Gamma_0(2)$ for $\dim X=4k\equiv 0(8)$.*

2.3. **Quantization of signature-expansion at cusp 0.** The modular curve of $\Gamma_0(2)$ has two cusps, namely 0 and ∞. Both $\varphi(X)$ and ε are written in the local coordinate q at the cusp ∞. If we write them in local coordinate $\tilde{q}$ at the cusp 0, then we have

Proposition 2.7.

$$\tilde{\varphi}(X)\cdot\tilde{\varepsilon}^{-k/2}=\tilde{q}^{-k/2}\cdot\hat{A}(X,\bigotimes_{n=2m+1}\Lambda_{-\tilde{q}^n}T_{\mathbb{C}}\otimes\bigotimes_{n=2m+2}S_{\tilde{q}^n}T_{\mathbb{C}}).$$

2.4. **The Witten genus.** Let $L\subset\mathbb{C}$ be a lattice, and let

$$\sigma_L(x)=x\cdot\prod_{\omega\in L\setminus\{0\}}\Big((1-\frac{x}{\omega})\exp(\frac{x}{\omega}+\frac{x^2}{2\omega^2})\Big).$$

Definition 2.8 (Witten genus)**.** *For a compact, orientable smooth manifold X^{4k} we define its Witten genus to be*

$$\varphi_W(X)=\Big(\prod_{i=1}^{2k}\frac{x_i}{\sigma_L(x_i)}\Big).$$

It turns out Witten genus is elliptic, and the power series $Q(x)=\frac{x}{\sigma_L(x)}$ has a beautiful product representation

$$Q(x)=\frac{x/2}{\sinh(x/2)}\cdot\prod_{n=1}^{\infty}\frac{(1-q^n)^2}{(1-q^ne^x)(1-q^ne^{-x})}e^{-G_2(\tau)\cdot x^2}.$$

Here $G_2(\tau)=\frac{1}{2}\sum_{\omega\in L'}\frac{1}{\omega^2}=-\frac{1}{24}+q+3q^2+\dots$ is the Eisenstein series of weight 2. If we use our standard notation $p(X)=\prod(1+x_i^2)$, then we obtain for Witten genus

Lemma 2.9.

$$\varphi_W(X)=\Big(\prod_{i=1}^{2k}\big(\frac{x_i/2}{\sinh(x_i/2)}\prod_{n=1}^{\infty}\frac{(1-q^n)^2}{(1-q^ne^{x_i})(1-q^ne^{-x_i})}\big)e^{-G_2(\tau)\cdot\sum_{i=1}^{2k}x_i^2}\Big)[X]$$

Corollary 2.10. *If X is a string manifold, then $p_1(X)=\sum x_i^2=0$, hence the Witten genus*

$$\varphi_W(X)=\hat{A}(X,\bigotimes_{n=1}^{\infty}S_{q^n}T_{\mathbb{C}})\cdot\prod_{n=1}^{\infty}(1-q^n)^{4k}$$

is a modular form of weight $2k$ with integral Fourier expansion.